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OH-PLIF Measurements for Micromixed Flames in a Multi-Injector Burner at Elevated Pressures and Temperatures

A multi-injector combustion system consisting of seven premix/micromix injectors is studied in the high-pressure optically accessible combustion rig (OACR), which is commissioned for the first time at the National Research Council Canada (NRC). The clustered design mimics the packaging expected in industrial applications where hundreds of such injectors will be needed to provide the desired power output. The pressure vessel and test facility allow for an investigation of the combustion system up to 922 K and 20 bar inlet temperature and pressure, respectively. It provides conditions closer to gas turbine operation to investigate the impact of pressure on micromixed flames and provide pressure scaling, such as for emissions and flame characteristics, to be used for extrapolating to full gas turbine load. Three of the five optical windows of the OACR are used in the current experimental configuration to monitor the flames with direct imaging and perform nonintrusive OH-planar laser-induced fluorescence (OH-PLIF), for the first time in this rig. The current work focuses on the combustion of pure hydrogen in micromixed flames to commission the added hydrogen capabilities of the test cell and the laser diagnostics. Globally-lean hydrogen flames operating at the same preheated air temperature, target flame temperature, and pressure drop across the injectors are investigated at pressures of 2.5, 5, 7, 10, and 15 bar to isolate the impact of pressure on combustion. Increasing pressure is found to significantly deteriorate the quality of raw OH-PLIF images, down to a mean signal-to-noise ratio (SNR) of 1.7 at 15 bar. Spatial filtering corrects for horizontal intensity striations believed to be caused by combustor cooling in the high-pressure environment, before a qualitative exponential relationship is used to correct for laser attenuation in the high-pressure environment and through multiple flames. The laser sheet slicing across the centerline axis of three inline nozzles mostly captures the main flame body in the turbulent wake of the micromix jets of the complex tridimensional structures. Little changes in flame structures are observed with increasing pressure, with the 15 bar condition moving slightly downstream from the nozzle exit. The results suggest that injector concept screening at lower pressure while maintaining the other operating conditions similar may be possible before pursuing additional elevated-pressure testing. [DOI: 10.1115/1.4069788]

Keywords: micromix, fuel flexibility, hydrogen, high-pressure, multi-injector array

1 Introduction

Hydrogen is a promising carbon-free alternative to decarbonize the energy and aviation sectors. In an effort to mitigate their environmental footprint, engine manufacturers are actively developing novel technology to safely and reliably burn hydrogen. While it offers high energy density, hydrogen cannot be used as a drop-in replacement for current hydrocarbon fuels in traditional, large, premixed combustion systems. Its high reactivity and preferential diffusion in premixed mixtures pose significant risks of flashback that may result in hardware damage or failure [1]. Technical innovations are therefore required to develop novel combustion systems, outside of the established design rules for hydrocarbon combustion, to enable the combustion of hydrogen in a safe and sustainable approach.

Micromix combustion technology is one avenue considered for the combustion of high-hydrogen-content and pure hydrogen fuels [2–4]. The micromix fueling strategy relies on multiple, miniaturized fuel injection orifices located close to the injector exit. This

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approach is intrinsically resistant to flashback as no reactive mixture flows through the injector body. If designed adequately, then the injected fuel, often introduced via jets-in-crossflow, rapidly mixes with the air and can help mitigate NO_x emissions through the miniaturized flames [3,5] and dampen the thermoacoustic response for high-hydrogen flames. Since micromixing is generally limited to high-hydrogen-content fuels only, recent developments aim to combine premixed and micromixed fuel injection strategies into a single compact injector body to offer fuel-flexible combustion systems able to stabilize flames from pure methane to pure hydrogen [4,6]. These geometries, demonstrated at atmospheric conditions, rely on complex internal fuel passages that have become possible thanks to recent advances in additive manufacturing (AM) [7].

Premixed/micromixed designs require multiple injectors, up to a few hundred in practical applications, to deliver the desired power output. Multinozzle configurations allow the study of flame-flame interactions in a geometry one step closer to that of practical systems. Few studies focus on confined clustered combustion systems [4,8–10] as these provide challenging environments to perform advanced laser-based diagnostics that often require unobstructed optical access to the flame bodies. Nonetheless, such experiments can provide key information on flame-flame interactions to be used for practical design optimization and combustion model validation. In previous design iterations of the injectors investigated in this work, significant flame coupling was observed not only between micromix jet-flames (i.e., the multiple fuel streams within a single injector tube), but also between neighboring injectors, leading to longitudinal pulsations in a confined environment, and also some rotational motion in the complex tridimensional flame topologies [10].

While atmospheric combustion rigs provide accessible platforms to perform fundamental research and preliminary design exploration for practical combustion systems, industrial gas turbines operate in preheat and high-pressure conditions. Better understanding the impact of boundary conditions on flame stabilization and dynamics is a necessary step to reach mature designs. That is especially crucial for hydrogen technology, which cannot rely on empirical design rules developed for hydrocarbon lean-premixed combustion. Such experiments often require significant engineering work where scientific investigations in high-pressure, optically accessible rigs are often limited by either operating pressures below 10 bar, preheating, or flow rates [11–13]. Optically-accessible, preheated, and high-pressure combustion rigs that can deliver industrial flow rates are seldom found and require significant investment to build and operate [14,15]. These can accelerate the development of novel technology by providing insight into combustion phenomena that are otherwise inaccessible.

The current work presents the high-pressure, optically accessible combustion rig (OACR) in the Test Cell #2 facility that was recently upgraded for hydrogen combustion at the National Research Council Canada (NRC). Nonintrusive laser diagnostics were implemented in the OACR for the first time to perform OH-planar laser-induced fluorescence (PLIF) measurements at elevated pressures. A 7-injector premix/micromix combustion system, improved via testing of various prototypes at atmospheric pressure [4,6,10,16,17], was installed in the pressure vessel for pure hydrogen experiments with preheated air from 2.5 to 15 bar. This first investigation at preheated and high-pressure conditions was primarily focused on the commissioning of the hydrogen facility and laser diagnostics. Stable 100% micromix H_2 flames were obtained for all the pressures studied. This work confirmed the operability of the upgraded hydrogen combustion facility and demonstrated the feasibility of the premix/micromix combustion strategy at operating conditions relevant for gas turbine operations. The dataset obtained provides insight into the micromix combustion strategy toward gas turbine implementation and the geometrical optimization of these unique injector features for practical applications.

2 Methodology

2.1 Optically Accessible Combustion Rig. The capabilities of the OACR are presented in Table 1, and its geometry is shown in Fig. 1. It is installed in the Class 1 div. 2 high-pressure, hydrogen combustion test cell #2 at the NRC Canada. The installation meets the ISO 9001:2015 standard for quality management. The rig consists of a high-pressure vessel in which optically accessible combustors can be installed. The OACR provides a flexible platform to perform nonintrusive diagnostics for single- and multi-injector combustion systems where the geometry under investigation only needs to mate with the mounting flanges of the pressure vessel.

The OACR consists of three distinct casings: air inlet section, cooling air section, and exhaust section, highlighted in Fig. 1(*a*). The air inlet receives the preheated air delivered at maximum conditions of 14.5 kg/s and 922 K. This air section is used for both combustion air and "hot" cooling air for the combustion system and liner to mimic the cooling found in industrial systems. The cooling, or medium temperature, section houses the combustion system. It provides low-temperature air up to 373 K, at a slightly higher pressure than the main air section ($\Delta P_{\text{mid}} = 0.3\%$), to cool the external components of the combustor and the exhaust tube. This section has an internal diameter of 0.45 m and a usable length of 1.2 m, providing a flexible platform to mount combustors of varying sizes. It is equipped with four optical access ports located at 0 deg, 90 deg, 150 deg, and 270 deg at the same longitudinal location (z). In this work, the 150 deg port is blanked and used for instrumentation and the spark ignition system. The other three openings are equipped with fused-silica windows (Spectrosil 2000) to provide visual access to the combustion process. The combustion and cooling air then flow into the exhaust section which is cooled with a water jacket. The combustion products are spray-cooled downstream of the exhaust before exiting through the facility stack.

The fuel inlet flange has four ports to accommodate primary (Py) and secondary (Sy) fuel streams for both natural gas and hydrogen. Two ports are used for each fuel stage to ensure a uniform fuel feed to the internal fuel manifold. The different fuel components are teed-in and mixed inside the pressure vessel before the fuel blend is delivered to the combustion system, as shown in Fig. 1(*b*). The facility can deliver up to 0.10 kg/s of hydrogen and 0.68 kg/s of natural gas. Hydrogen is supplied via a dedicated outdoor Synthetic Fuel Facility that comprises four bays, each of which can each accommodate a compressed hydrogen jumbo tube trailer (Air Liquide, capacity of 4000 standard m^3 and pressure of 2400 psi), and a regulation pad to gradually decrease the hydrogen supply pressure to ~650 psi before delivery to the test cell.

Air and fuel flows are measured with appropriately sized orifice plates. Calibration of the flow measurement instruments was performed at the beginning of the project and validated regularly with a reference Coriolis mass flowmeter calibrated by the supplier. Although not all are listed in Fig. 1, the rig is fully instrumented along the pressure vessel and the combustor with dozens of thermocouples, differential pressure transducers, dynamic pressure transducers (CP211, Meggit ©), and absolute pressure measurements. The operation of the rig is controlled using a combination of LabVIEW and Lookout.

Emissions measurements are sampled by a multihole probe located in the exhaust section, shown in Fig. 1(*b*). Sampling is performed following the ARP1256D standards [18] when the flames stabilize and are averaged over 60 s for steady-state measurements.

Table 1 Optically accessible combustion rig

Air flow	Main air	14.5 kg/s	32 lb/s
	Cooling air	6.6 kg/s	14 lb/s
Air temperature	Main air	922 K	1200 F
	Cooling air	373 K	212 F
Pressure absolute		2068 kPa 20.68 bar	300 psi
Hydrogen flow		0.10 kg/s	0.22 lb/s
Natural gas flow		0.68 kg/s	1.5 lb/s
Test section	Diameter	45 cm	18 in
	Length	120 cm	47 in
Optical access	Midsection	4 (0°, 90°, 135°, 270°)	
	Exhaust	1	

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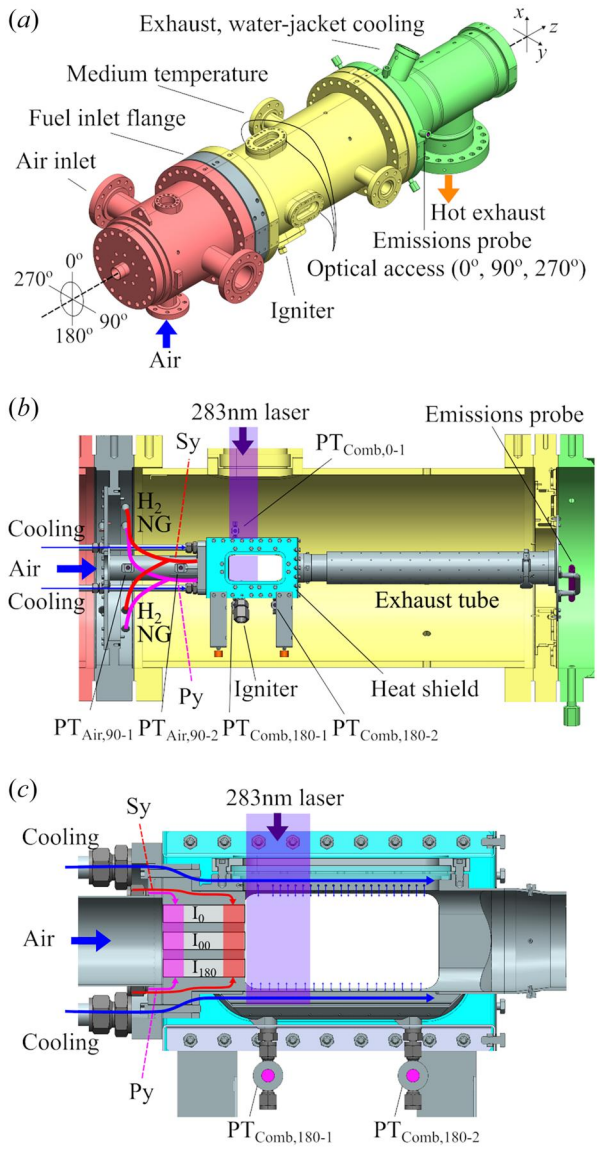

Fig. 1 High-pressure, optically-accessible combustion rig (OACR) with partial instrumentation: (*a*) external pressure vessel, (*b*) combustion rig housed in the medium temperature section, and (*c*) cross section of the combustion system at the location of the laser sheet used for diagnostics. Generic flow paths are provided for the main hot air used for combustion and cooling, hydrogen fuel lines, and natural gas fuel lines.

While the system can provide measurements of unburnt hydrocarbons, CO, and CO_2, only the NO_x (NO and NO_2) emissions are relevant for the current study burning pure hydrogen.

2.2 Micromix Combustion System. The micromix combustion system consists of three major components: the injector cassette, the optically accessible combustor, and the heat shield around the combustor. The cassette and combustor are components made of Inconel 625 through a monolithic AM approach. They were printed on EOS M290 machines using laser powder bed fusion. The AM allows for the complex integration of the fueling and cooling passages through the injector cassette and the cooling, especially for the windows, of the combustor. Conventional machining is carried out after printing to prepare surfaces that mate with other components, such as the window frames of the combustor.

The injector cassette contains the seven injectors and also acts as the fuel plenum for both the primary (premixed) and secondary (micromixed) fueling strategies. The internal fuel plenum for each fuel stage is fed via two pairs of inlets located in opposite corners. A complex system of internal channels then uniformly distributes the fuel to the injection points on each injector, at the respective fueling zones identified in Fig. 1(*c*). Primary fuel injections occur close to the injector inlet and are sufficiently far upstream to achieve premixed conditions at the nozzle exit. Secondary fuel injections are performed as jets-in-crossflow near the injector exit and correspond to the micromix approach. The fuel split, defined as the micromix fuel ratio (MFR) in (1) with volumetric flow rates ($\dot{V}$), can be adjusted during experiments. In this work, however, conditions at MFR100 corresponding to 100% micromixed combustion are maintained across the entire dataset.

$$\mathrm{MFR} = \frac{\dot{V}_{\mathrm{Sy}}}{\dot{V}_{\mathrm{Py}} + \dot{V}_{\mathrm{Sy}}} \tag{1}$$

The orientation of the injectors is shown in Fig. 2. Labeling of the injectors is consistent with their angular positions relative to the coordinate system of the OACR. The three inline nozzles I_0, I_{00}, and I_{180} are the ones investigated with laser diagnostics. The cassette is equipped with multiple type-N thermocouples located on the back side of the dump plane and embedded in the injectors to measure metal temperatures and capture potential flashback or flame anchoring that might lead to hardware damage.

The preheated air flows through four tubes from the air inlet section of the OACR into the combustor to provide "hot" cooling air for the windows and the laser beam dump of the combustor. Partial window cooling air paths are presented in Fig. 1(*c*) and mitigate the thermal stresses in the brittle fused-silica windows during operation. The combustor has optical access on three sides at 0, 90, and 270 degrees, while the 180-degree side includes a laser beam dump, as shown in Fig. 2 to minimize the reflections back into the analysis region of interest for nonintrusive diagnostics. The windowless side is instrumented with six type-N thermocouples to measure metal temperature profiles along the combustor length, and provides space for the ignition system.

Five dynamic pressure transducers (CP211) are mounted on the combustion system. They are identified in Figs. 1(*b*), 1(*c*), and 2, which provide the location and orientation of each instrument. Two pressure sensors ($PT_{\mathrm{Air,90\text{-}1and\text{-}2}}$) are located upstream of the flames in the air plenum, and three in the combustion chamber($PT_{\mathrm{Comb,0\text{-}1}}$ and $PT_{\mathrm{Comb,180\text{-}1and\text{-}2}}$). Pressure signals are acquired at a 50 kHz sampling frequency, and the acoustics module also acquires the scientific camera gate to allow for phase-average imaging [10].

2.3 Laser Diagnostics. The hydrogen-rated, high-pressure test cell is a class 1 div. 2 hazardous location. As such, electrical equipment must be either intrinsically safe, sealed, or purged/pressurized. The classification further complicates the diagnostics setup as the lasers are located outside of the test cell, producing a laser path 8.5 m long from the dye laser head to the experimental region of interest, and the camera systems have to be housed in a purged, positive-pressure enclosure. The photograph of the OACR in Fig. 3(*a*) highlights the end of the laser path close to the pressure vessel. In this configuration, an explosion-proof CCTV camera is mounted on one optically accessible side for safety reasons during the commissioning of the hydrogen and laser systems. The scientific cameras are installed facing the opposing window in a purged box (Fig. 1(*b*)) that is purged with dry air and maintained at a positive pressure of approximately 80 psig. The purge box is equipped with a single fused-silica window.

The second harmonic of a Spectra-Physics Nd:YAG laser (Quanta-Ray Pro-250-10H) pumps a wavelength-tunable dye laser (Sirah, Precision Scan) using a solution of rhodamine 6G. The dye laser is adjusted at a wavelength of 282.94 nm and produces an ultraviolet beam with an average output energy of ~8.5 mJ/pulse at the exit of the laser housing. Laser line mirrors then direct the beam through the wall of the high-pressure test cell and above the OACR. Sheet forming optics are installed on the horizontal top rail crossing

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above the OACR and adjusted such that the thin laser sheet waist is centered slightly downstream from the outlets of the injectors (Fig. 1(*c*)). The resultant laser sheet spans ~4 injector diameters, *D*. In addition to the long optical path and generally larger length scales than those found in atmospheric combustion rigs, the alignment is further complicated by the fact that it is performed when the rig is opened and cold. When preheated, the OACR can thermally expand in the longitudinal *z*-direction by as much as 50 mm at the exhaust and by approximately 1.4*D* at the location of the optical access. Therefore, the procedure for laser alignment factors in the thermal expansion to optimize the location of the beam relative to the dump plane of the cassette.

OH-Planar Laser-Induced Fluorescence images are acquired using an sCMOS camera attached to an IRO (LaVision IRO). The fluorescence signal is collected by the IRO, equipped with a bandpass filter (320±20 nm) on a UV-Nikkor lens (105 mm), with a gate time of 200 ns and a gain of 70% of maximum. In these experiments, the aperture is adjusted to f/8 to slightly increase the depth of field to allow for slight misalignment as the rig may shift over the course of an entire day of testing. The smaller aperture also mitigates the amount of realignment and refocusing that might otherwise be needed once the OACR is completely closed. Raw images are corrected for pulse-to-pulse laser energy variations measured by a photodiode close to the laser exit and background noise. Spatial calibration of the camera is performed with a 3D LaVision calibration plate (Type 20). After cropping, the experimental field of view captures a region of interest spanning approximately 5.3*D* × 3.5*D*. The camera and laser system are triggered synchronously at a frequency of 10 Hz using a LaVision timing unit (PTU-X), and 600 images are acquired per dataset.

2.4 Operating Conditions. Experiments in the current study maintained similar operating conditions, except for the inlet air pressure, which varied from 2.5 bar to 15 bar. Measurements are performed for 100% hydrogen fuel burning in a full micromix fuel injection strategy (MFR100). The same air inlet temperature and target adiabatic flame temperature are maintained for the different pressure conditions to achieve globally lean reactant mixtures. The target, constant, adiabatic flame temperature replicated primary zone temperatures of interest for gas turbine applications. Inlet flow rates were adjusted to maintain the same pressure drop across the injectors for all experiments.

Measurements of effective flow area were carried out before each test day to ensure the integrity of the injectors and to calculate the flow split, and therefore the zone temperatures. Post-test inspections were conducted after each test day to identify potential discoloration or damage to the hardware and ensure the validity of comparisons between different days. The data reported in this work have been acquired over multiple days with no discrepancies observed in validation metrics.

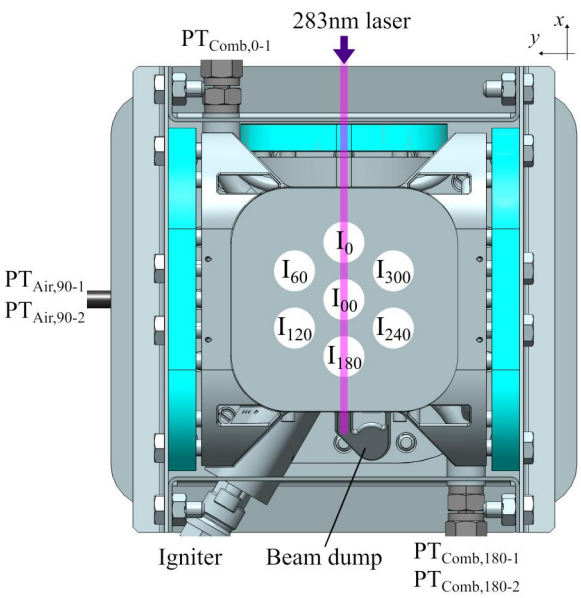

Fig. 2 Upstream view of the combustion system

Stable micromixed flames were obtained for pure hydrogen at pressures of 2.5, 5, 7, 10, and 15 bar. During testing, the 7-injector cassette could sustain the target flame temperature up to 15 bar without damage to the hardware under nominal operations. Several high-pressure, 100% hydrogen relight tests were performed to assess the robustness of the combustion system and the OACR to hydrogen ignition events. While the ignition process generally sheds particles and small debris accumulated in the upstream pressure vessel and the airlines that can coat the windows of the combustor, no window failures were observed during testing.

3 Results

3.1 100% Hydrogen Micromix Combustion. The direct visualization of the flames investigated is shown in Fig. 4. The images were captured by a Nikon D750 DSLR camera using the same acquisition settings at all operating pressures. The thermal power of the combustion system increases with pressure, resulting in the most luminous flames being captured at 15 bar. The absence of light in the pressure vessel and the low-light condition in the test cell allow the hydrogen flames to be visualized, although it is difficult to distinguish the flames coming from individual injectors, given the angle of view of the camera relative to the orientation of the injectors in the cassette. In these images, the white hydrogen flames appear to be fully lifted from the injectors and the products in the postflame regions are indicated by the red/orange glow downstream of the hydrogen flames. As thermal power increases, the combustor also experiences higher operating temperatures. This is evident from the direct imaging as the sides of the window frames start glowing. During testing, the metal temperatures of the test section were monitored closely to ensure that they did not initiate the automated fuel trip sequence that shuts down fuel supply to preserve the combustion system integrity. Red spots can also be observed in the more luminous cases downstream of the flames. These correspond to particles that have stuck, and in some cases fused, to the windows during ignition. These spots were removed as best as possible between tests to maximize the transmissivity for optical diagnostics.

During testing, several attempts were made to transition from a micromixed condition to a partially micromixed condition in an effort to approach premixed combustion. For pure hydrogen fuel, these resulted in flashbacks at high micromix fuel ratios, typically above MFR85, even for globally leaner conditions with increased inlet velocity. These attempts highlight the benefits of micromixing for the premix/micromix design to increase the fuel flexibility of the injectors to safely stabilize pure hydrogen flames.

Normalized NO_x emissions are shown in Fig. 5 for 100% H_2 (MFR100) conditions with increasing pressure. NO_x measurements are obtained in concentration units and corrected to a dry, 15%O_2 standard (15%O_2 ppmvd). They are then normalized by the 15 bar condition. The data points shown here correspond to an average of 60 s acquired at the target condition. From these experiments conducted at the same target flame temperature and pressure drop across the injectors, it is possible to extract the pressure exponent needed to develop a surrogate model that can be used to predict emissions at higher pressure ratios. The micromix strategy provides promising NO_x emissions for a first design iteration at elevated pressure, and the exponential trend achieves an $R^2 = 0.987$ to estimate emissions at higher operating pressures found in some industrial systems. Further modification and optimization of the design are expected to yield lower NO_x emissions in future designs.

3.2 High-Pressure OH-Planar Laser-Induced Fluorescence Correction. Raw OH-planar laser-induced fluorescence (OH-PLIF) images at 5, 10, and 15 bar (Fig. 6) show the flames

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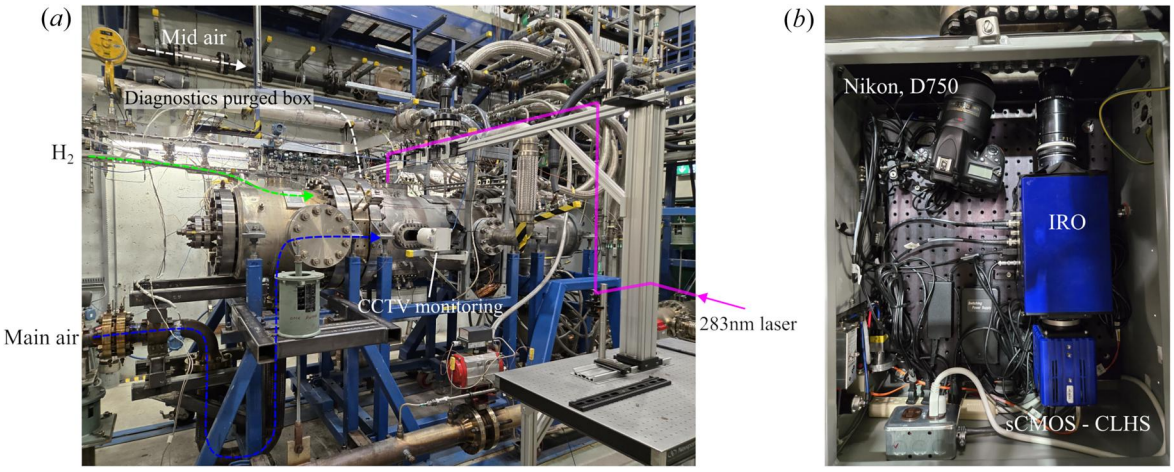

Fig. 3 (*a*) OACR installed in test cell #2 at the National Research Council Canada. The partial laser path is highlighted in magenta. (*b*) Purged box containing the camera equipment used for direct visualization and laser diagnostics.

corresponding to the three inline injectors I_0, I_{00}, and I_{180}. In these images, the laser propagates from the right of the frame to the left ($x = 2D \rightarrow -2D$) and subsequently traverses the three flame regions. For these conditions, the raw images generally show a compact flame downstream from each injector that exhibits higher laser-induced fluorescence (LIF) intensity, and unburnt and cold air streams between the injectors, identified by the absence of signal. The raw images clearly show the challenges of performing fluorescence at elevated pressures, as highlighted in various other works [11,12,19]. Three observations are made from the raw OH-PLIF images.

(1) The mean signal-to-noise ratio (SNR) for single-shot decreases significantly with pressure. Including the three flames captured in the field of view, the SNR decreases from 3.1, 2.1, and 1.7 as the pressure increases from 5, 10, to 15 bar.
(2) There is a significant variation in signal intensity from the right to the left side of the raw images. At 5 bar, the SNR corresponding to injectors I_0, I_{00}, and I_{180} has values of 4.8, 2.7, and 1.5, respectively. This laser energy attenuation is generally negligible in single burners at atmospheric pressure. It is, however, typically present in multi-nozzle combustion systems, even at atmospheric conditions [4,9] and also in single burner experiments performed at elevated pressures [11,12,19].
(3) Horizontal striations become visible as pressure increases beyond 5 bar. These are most prominent at 15 bars and appear to be regularly spaced. During testing, moderate (5 bar) and high (10 bar) pressure conditions were sequentially investigated, and it was confirmed that the striations are negligible in the lower pressure condition, even if they were present in the previous test point. This indicates that it is not a systematic bias created by a damaged or dirty window.

While the first two observations are unavoidable for LIF at elevated pressures, it is suspected by the authors that the striations result from design choices in the combustor that will be explained later. The combination of these three phenomena in a multi-injector combustion system complicate the analysis of the acquired data. This is even more true for micromix combustion systems that have multiple distinct fuel orifices located near the injector outlets, as highlighted in the previous 3D, OH-LIF reconstruction of similar burners [10]. Nonetheless, in the linear-LIF regime, which is estimated to be the case here, signal attenuation is independent of the initial laser energy, contrary to saturated LIF. These effects can be partially corrected in instantaneous images, although they will lead to increased noise in the leftmost flames, whose SNR is the lowest among the three injectors.

Raw images are first corrected for shot-to-shot laser energy fluctuations before the background noise is removed to obtain the OH signal only. The images are then corrected for the spatial, longitudinal (z), fluctuations that appear as horizontal striations. The intensity variation appears to be fairly consistent along the three flames investigated, with a roughly regular spacing. While the exact

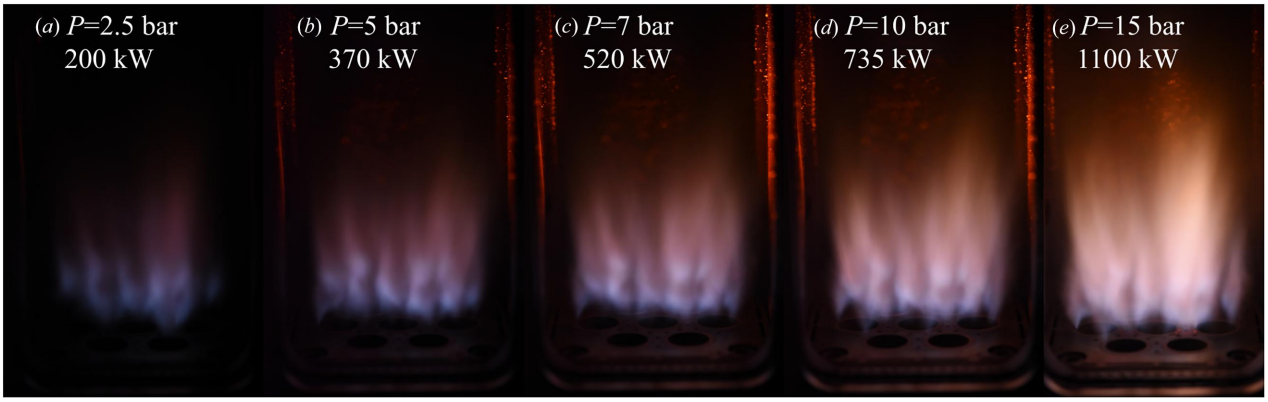

Fig. 4 Direct flame images acquired with a Nikon D750 DLSR camera. Camera settings are maintained for the different operating pressures at f/5.6, 1/320 s exposure time, and ISO-3200. The same pressure drop across the injectors and target adiabatic temperature were used for all cases. The increased luminosity results from the increased thermal power as operating pressure goes from (*a*) 2.5 bar up to (*e*) 15 bar.

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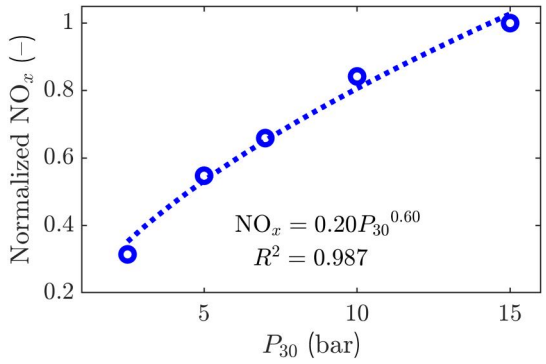

Fig. 5 Normalized NO_x emissions

cause of the observed behavior has yet to be determined, the authors hypothesize that it is a combined effect of the elevated pressure and the window cooling that introduces a regular pattern in the incoming laser sheet. The combustor window through which the laser sheet is entering is cooled by impinging jets issued from the window frame of the combustor. While this cooling is active at all pressures, the authors suspect that temperature/density gradients between cooling air and recirculated products in the vicinity of the window where the laser sheet enters the combustor create a roughly structured pattern of flow pockets with different refractive indexes. This pattern could effectively act similarly to an array of beam shaping optics, with the hot and cold flow pockets acting as microlenses and leading to the horizontal pattern observed. Smaller deviations of this pattern can be observed through the inline flames as the laser sheet traverses regions that can be made of hot product, cold air, and mixes of both across the three injectors. Future testing is planned that will include a different configuration of the cooling strategy for the laser inlet window to validate this hypothesis. Another monolithic additively-manufactured combustor is currently being redesigned to mitigate the amount of cooling air in the region, where laser diagnostics is performed while maintaining material integrity at the target operating condition.

With the current geometry, spatial filtering is applied to correct the striations observed in instantaneous raw images along the laser path. It is demonstrated in Fig. 7, starting with the same raw image shown in Fig. 6(*b*) for the 10 bar pressure condition. A moving average is first performed along the x-direction with a narrow longitudinal window (along z) in Fig. 7(*a*). This further highlights the intensity strata observed previously. The raw image is also processed using a Gaussian smoothing with a narrow x window along the z-axis to blend, or average, the local OH-PLIF intensity in Fig. 7(*b*). This effectively not only removes the horizontal striations but also greatly blurs the image and removes the smaller flow features captured in the raw image. The ratio of these two is shown in Fig. 7(*c*) for reactive pixels in the frame identified with a wide OH mask. The local fluctuation mask captures the moving average intensity variations with respect to the smoothed image. It can be multiplied by the raw image (Fig. 6(*b*)) to correct for the horizontal striations. As mentioned previously, these lines of intensity variations appear to be very regular and mostly occur along the laser sheet. In a few instances, they appear to slightly deflect from the horizontal as the laser sheet travels through the internozzle region where thermal and density gradients can lead to different diffractive indices.

The processed image can then be corrected for the laser attenuation caused by OH molecules along the laser beam path in the high-pressure environment across the different flames. This correction is applied to instantaneous images, making use of the assumption that the different injectors are producing similar flames with equivalent averaged properties, for example, temperature or OH concentration. Compensating for the laser attenuation at high pressure has been an issue discussed in other work and tackled with various levels of complexity [11,12,19]. In this work, the attenuation correction remains relatively simple and does not fully correct the attenuation observed. It is meant to rebalance the intensity of OH-PLIF to provide datasets more apt for systematic analysis with combustion modeling. The approach is graphically represented in Fig. 8 for the same 10 bar pressure condition used previously. Under the linear LIF regime, the laser attenuation should be independent of the initial laser energy. The LIF intensity for each assumed identical flame then becomes an indirect measurement of the laser attenuation, and Beer-Lambert's Law [20] can then be applied to estimate LIF intensity reduction along the laser path for the inline flames using exponential fits. As stated previously, this is not a rigorous approach but rather a qualitative means to partially correct the attenuation observed in these high-pressure flames. In Fig. 8, the 2D OH signal intensity from an interrogation window along the z direction, here the segment between 0.7 and 1.4 z/D, is projected on the x-axis. Piecewise exponential fits are then applied over sections where flames are detected with an SNR threshold. The multiple fits from the different interrogation windows are then interpolated and smoothed over the entire field of view to generate a 2D map of laser attenuation that can be used to partially correct the image.

The results of the different image processing steps are shown in Fig. 9 for the 10 bar pressure condition. Each image is also processed using a 3 × 3 median filter to provide a systematic comparison between the intermediate steps. Filtering with a larger window would help remove additional speckling and noise, if needed. The raw image in Fig. 9(*a*) has an average SNR of 2.1, which is mainly affected by the central and left injectors. While the leftmost flame can hardly be distinguished in the raw image, the processing approach significantly improves the contrast and produces a

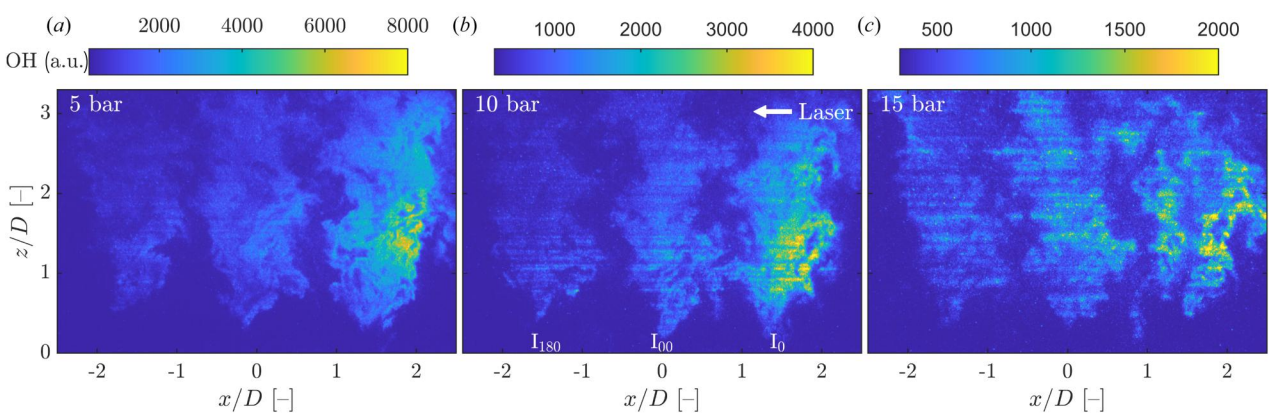

Fig. 6 Raw OH-PLIF images captured at (*a*) 5 bar, (*b*) 10 bar, and (*c*) 15 bar. Note the maximum OH intensity decreases with pressure.

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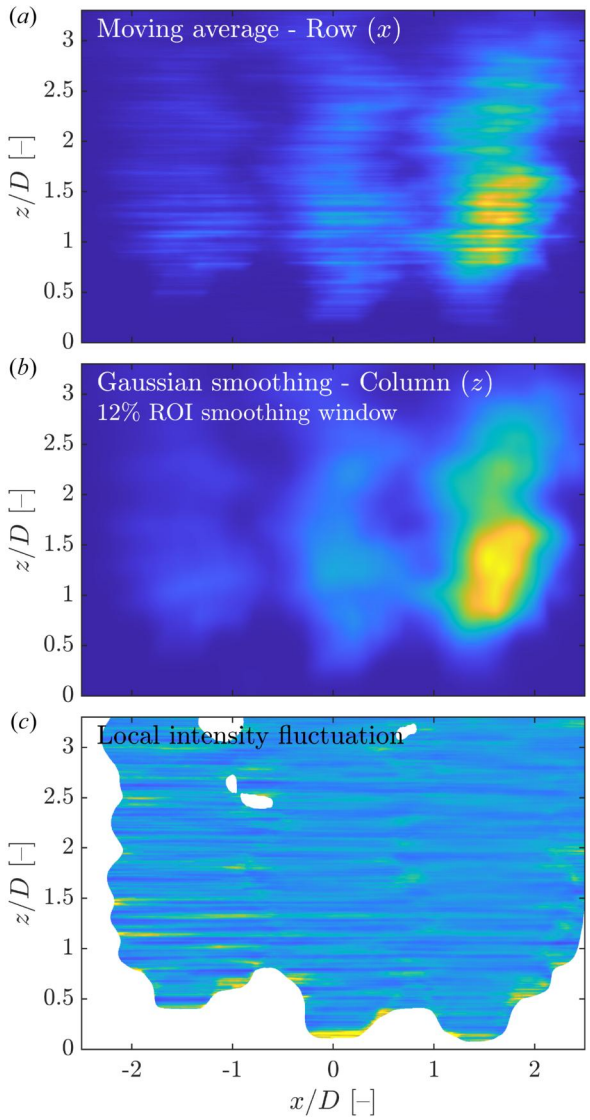

Fig. 7 Example of spatial filtering performed for the 10 bar condition. (*a*) Moving average performed in the *x*-direction further highlights the striations observed in the raw images. (*b*) Gaussian smoothing applied in the *z*-direction with a 0.4*D* moving window. (*c*) Local intensity fluctuations shown for a wide flame mask. The local intensity fluctuation mask is then used to spatially correct raw images.

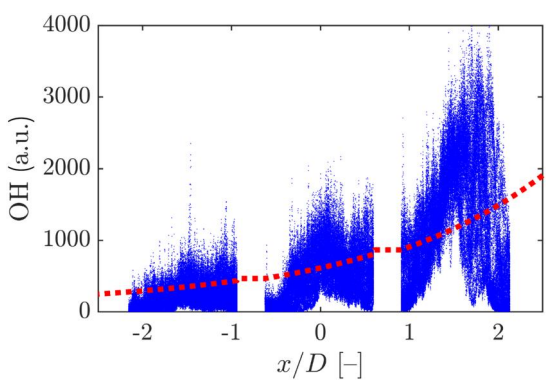

Fig. 8 Projection of OH intensity between 0.7–1.4z/D for the 10 bar condition. A piecewise, exponential fit is performed (–) to approximate the Beer-Lambert laser attenuation through the inline flames in the laser path. The fit obtained is then used to rebalance the OH intensity fields.

corrected image that is qualitatively more representative of the flames and OH-PLIF intensity expected. Using the spatial filtering mask, the horizontal striations can be mostly removed in Fig. 9(*c*) without compromising spatial resolution. Contrary to the large Gaussian smoothed average used to derive the mask, fine structures are retained in the processed images. The attenuation correction then increases the OH signal intensity along the laser path in Fig. 9(*d*). While this operation performs adequately for the central and right flames, additional noise is also being introduced in the final image for the left flame. More granularity, or speckling, is observed, which can be mitigated if a larger median filter is used to visualize the final OH-PLIF fields. This is expected as the last flame on the laser path can have SNR between 1.3 and 2, depending on pressure. That makes it difficult to clearly distinguish the signal from the background noise, especially at the weaker edges of the flames.

3.3 Flame Characterization With OH-Planar Laser-Induced Fluorescence. The corrected OH-PLIF images are presented in Fig. 10 for instantaneous shots and mean fields at 5, 10, and 15 bar. These flames were selected initially because of their relatively quiet, stable condition. Phase-averaging is not performed as no significant dominant frequency, nor a pressure limit cycle, is attained. While the image processing approach does significantly improve the contrast from the raw images, the mean OH fields clearly show that the piecewise exponential fit used to correct for laser attenuation through the three flames only partially corrects laser attenuation across the flame. The left flame still exhibits lower intensities compared to the other elements of the multi-injector combustor. In addition, window fouling starts being perceptible in mean fields where particles have fused onto the optical access windows.

Interpretation of these results remains challenging as the highly turbulent micromixed flames have no clear structure and are generally not axisymmetric [10]. With the laser sheet aligned with the injector centerline axes, the fluorescence images capture the bulk of the flame body downstream of the injector core. When comparing the PLIF images (Fig. 10) with the direct flame images (Fig. 4), it appears that the distributed lee-stabilized flames from the jets-in-crossflow are mostly not captured by the laser plane slicing through the combustion system. Three distinct flames with limited flame merging between neighboring injectors are observed. For pure micromixed combustion, the localized fuel injections mitigate the migration of premixed fuel-air pockets, which can burn between injectors. In some of the frames, such as for the 10 and 15 bar conditions shown in Figs. 10(*b*) and 10(*c*) for the right injector, part of the lee-stabilized hydrogen jet flame is hooking back upstream toward the injector outer diameter. Given the complex tridimensional flame structures and the clocking, or orientation of the micromix fuel injections within each injector, it appears that the anchored lee-stabilized part of the flames seldom appear in the field of view. The flame anchoring on the outer diameter of the injector can lead to high heat transfer to the hardware and locally creates hot spots on the injector heat shield, causing the hardware to be glowing hot. The metal temperature then becomes one operational limit at 15 bar for this geometry. Some of those hot spots can be captured with the DSLR camera during testing, as shown in Fig. 4(*e*) for the 15 bar condition between injectors I_{00}, I_0, and I_{300} on the right of the frame. Post-test inspection showed a discoloration of the metal, indicating that the hot spots were localized and firmly anchored in the flow for the current micromixed experiments. The pattern is repeated around the central injector spaced by 120 deg between the injector trios I_{00}, I_{180}, I_{240} and I_{00}, I_{60}, I_{120}. Given the complex tridimensional nature of the micromixed flames, appropriate orientation and spacing of the injectors have to be optimized along with cooling strategies to mitigate the high heat flux to the hardware. The location of these hot spots has been confirmed numerically after the experiments using preliminary computational fluid dynamic analyses. These findings provide an example where the synergistic impact of optically accessible combustion rigs with combustion modeling can rapidly identify undesired behavior.

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The overall flame topology remains similar for the micromixed flames as pressure is increased. This fueling strategy does not appear to be experiencing thermodiffussive instabilities and flame front thinning that is generally expected from globally-lean premixed configurations. For all the pressures investigated here, the flames are mostly lifted and present a broad OH reaction zone. The flame liftoff becomes more apparent as pressure increases to 15 bar and the leading edge of the core of the flame moves further downstream. The instantaneous OH-PLIF images show the downstream displacement with pressure, especially for the right injector (Fig. 10(*c*)), where the core of the flame now reaches approximately $0.5D$ downstream from the injector outlet. This is captured in the average OH field (Fig. 10(*f*)) as the flames more rarely move upstream the $z = [0, 0.5D]$ for all the nozzles. Flame centroids are also moving further downstream at the 15 bar condition. The overall qualitatively small changes in flame topology suggest that if other operating conditions can be reproduced at low-pressure conditions, such as inlet temperature, inlet velocity, and jet momentum flux ratios, promising design identification may be performed in atmospheric facilities at significantly reduced costs compared to high-pressure experiments. Further investigations of quantitative metrics should be performed to ensure that the micromix flame topology can scale to pressure.

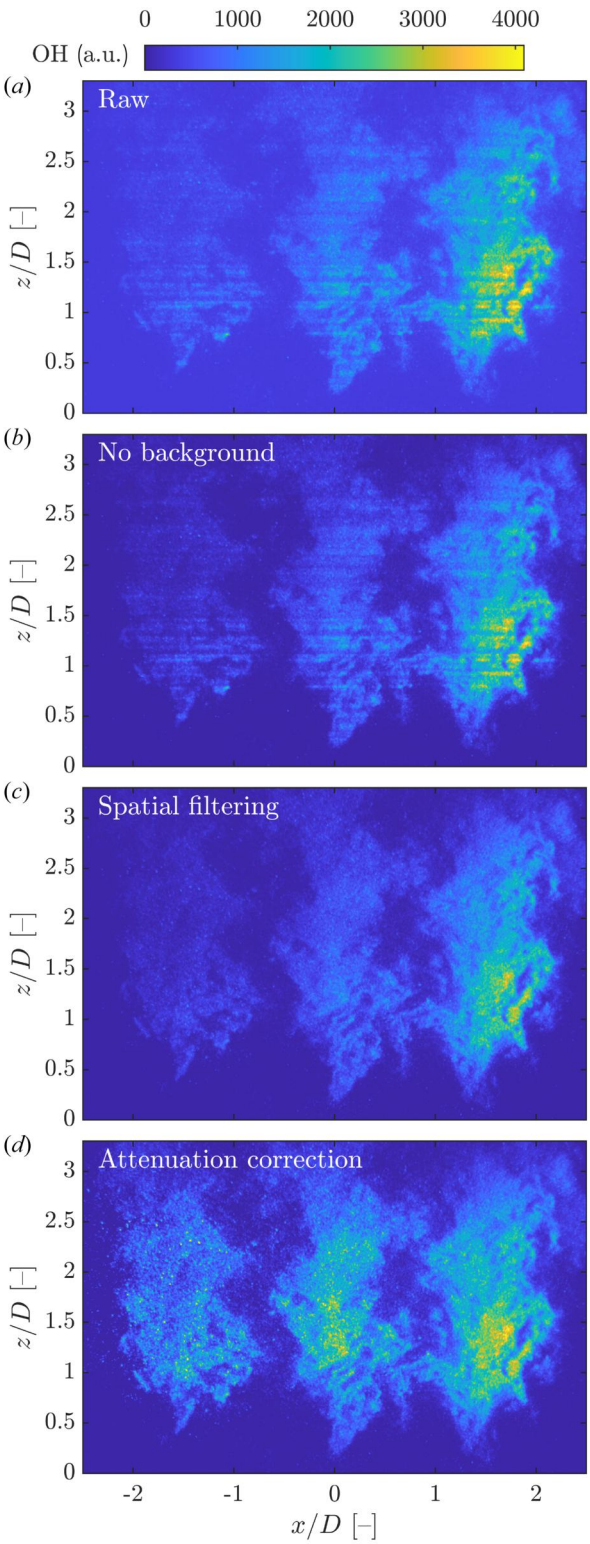

Fig. 9 Image correction process applied to the 10 bar condition and presented with a 3 × 3 median filter for direct comparison between the processing steps

Approaching gas turbine operating conditions offers unique datasets that can be used to further understand the performance of different injector designs toward higher technology readiness integration. It is a necessary milestone in new combustion system development to identify pressure scaling, stable operating regions, and emissions trends. The OACR offers optical access to leverage the planned high-pressure testing needed for industrial development to provide additional information on the flame structures that are otherwise typically modeled only. It offers insight into the combustion process that can then be used to better understand the hardware being tested and also provide complete validation datasets for combustion model development. Moving toward high-hydrogen combustion, industrial computational fluid dynamic best practices have to be modified to account for the different physical properties of hydrogen compared to natural gas or other hydrocarbons. This dataset will be used in subsequent work to provide a baseline for design improvement and will infer changes needed to improve the premix/micromix designs.

After commissioning the OACR and hydrogen facility, additional work is planned on this injector configuration to expand the scope of testing to encompass fuel blends with natural gas and to investigate the impact of MFR on the hybrid premix/micromix combustion system. The entire body of work should then be used for design iterations that will be tested in the OACR in upcoming experimental campaigns. The modular rig allows for quick turnover between combustion systems to do advanced screening between interesting geometries to better understand the stabilization mechanisms between promising premix/micromix injectors.

4 Conclusion

The high-pressure OACR is presented for the first time. It is used to commission the nonintrusive laser diagnostics and the hydrogen fueling upgrades recently completed in the Test Cell #2 facility at the National Research Council Canada. The adaptable test section of the pressure vessel is equipped with a new 7-injector combustion system relying on the combination of premixed and micromixed staged fueling strategies that offers wide fuel flexibility from current hydrocarbons to pure hydrogen. In this work, the combustion of pure hydrogen in the micromix combustion mode is investigated with preheated air at elevated pressures. The optical access is used to perform direct flame imaging with a Nikon DSLR camera and to perform OH-PLIF diagnostics at elevated pressures.

The hydrogen capability of the 7-injector cassette is confirmed at pressures of 2.5, 5, 7, 10, and 15 bar by achieving stable micromix flames at each condition. Steady-state acoustics and emissions measurements are acquired at each data point and confirm the potential of the premix/micromix approach at operating conditions closer to gas turbine applications. Since carbon emissions are nonexistent for pure hydrogen combustion, only a NO_x pressure scaling is extracted from the emissions measurements to be used to estimate pollutant formation at higher operating pressures.

Laser diagnostics are performed for three inline injectors with the laser sheet centered above their nozzle centerline axes. The results

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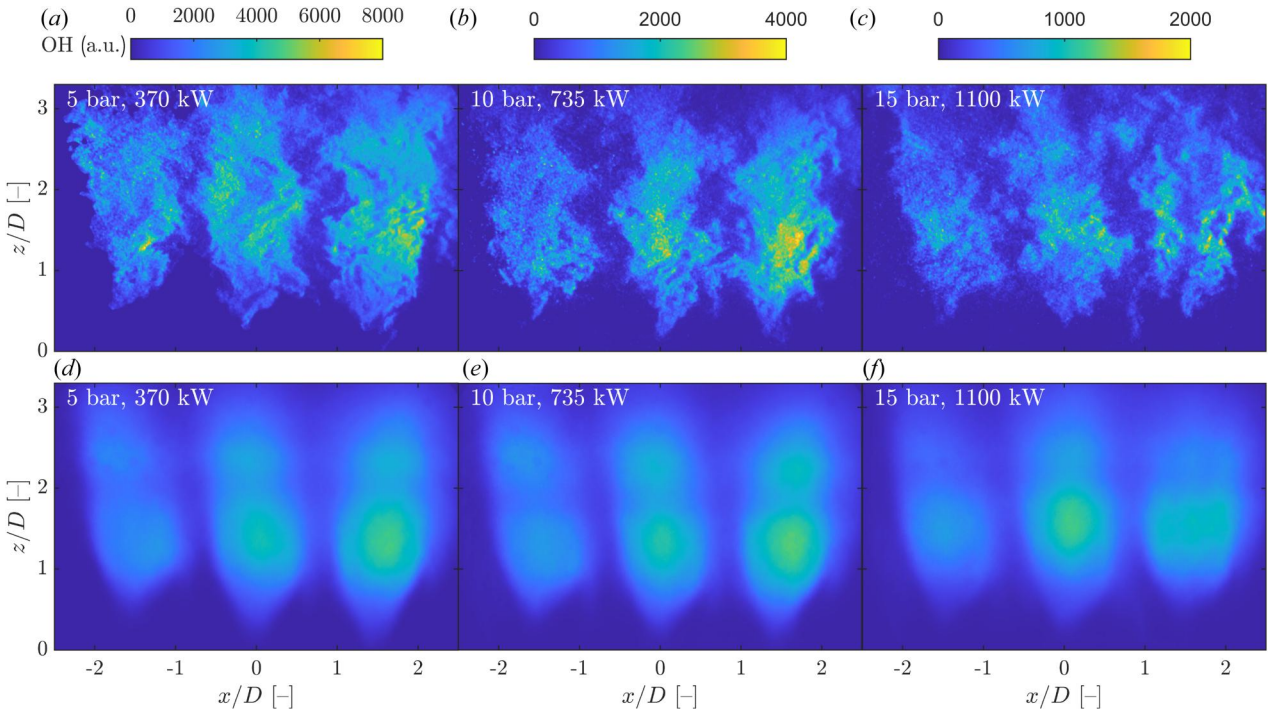

Fig. 10 (*a*)–(*c*) Instantaneous OH-PLIF fields corrected for spatial variations and attenuation for increasing pressures. (*d*)–(*f*) Mean OH-PLIF fields. Note that the OH-PLIF intensity scale is different at increasing pressures, but consistent for instantaneous and mean fields.

for the 5, 10, and 15 bar pressure conditions highlight the challenges of performing optical diagnostics at elevated pressure conditions with multiple-injector clusters. Raw OH-PLIF images present significant SNR reduction with increasing pressure and along the inline injectors because of laser attenuation. In addition, striations appear in the images, which are suspected to be caused by a combination of variations in the refractive index of the hot combustion products and the relatively colder cooling air jets, combined with the impact of pressure of gas densities. Qualitative approaches are proposed to correct for spatial inhomogeneities in the laser profile and for the laser attenuation through multiple flames. The corrected images provide qualitatively better contrast for all the flames to be compared between different pressure conditions.

The OH-PLIF images mostly capture the lifted turbulent micromixed flames in the wake of the fuel injection. Given the complex tridimensional flame topology and location of the laser sheet with respect to the localized micromixed fuel injection, only a portion of the lee-stabilized hydrogen jet flame anchored on the outer injector diameter is captured in a limited number of frames. The micromixed flame structure appears mostly unchanged with increasing pressure, suggesting that experiments at lower pressure might be pursued for preliminary characterization, if other operating conditions can be maintained. As pressure is increased to 15 bar, the flames lift slightly further downstream of the injector exit and create recirculating hot pockets on the injector dump plane. This leads to higher metal temperature that can limit the operational range.

This first investigation using the OACR shows the potential of high-pressure optical diagnostics to accelerate the development of hydrogen technology. The datasets obtained provide information to better understand where and how flames stabilize in the elevated temperature and pressure environment closer to gas turbine operation. They can also be used to validate combustion modeling with direct measurements of the combustion phenomena to help converge toward an optimal injector design. Leveraging already planned high-pressure testing necessary for novel multi-injector systems, individual injectors, or sector rigs, the pressure vessel can provide additional and complementary information to traditional rig measurements to streamline the design process toward higher technology readiness levels and fully utilize the costs of high-pressure testing. With the commissioning completed, further investigations on the effects of fuel split and fuel composition on the premix/micromix combustion are planned, along with the testing of design iterations.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

AM = additive manufacturing
D = nozzle diameter (m)
DP = differential pressure
HP = high pressure

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I = injector
IRO = intensified relay optics
MFR = micromix-fuel ratio
NG = natural gas
OACR = optically-accessible combustion rig
P = pressure (kPa)
PLIF = planar laser-induced fluorescence
PT = pressure transducer
PT = pressure transducer
P_y = primary fuel channel
S_y = secondary fuel channel
SNR = signal-to-noise ratio
$\dot{V}$ = volumetric flow rate (SLPM)

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